

Cerebrospinal fluid and the blood-brain barrier

The written notes for the cerebrospinal fluid and the blood-brain barrier walkthrough. Week 5 competency 6.

Step by step: where the signal travels

Where cerebrospinal fluid travels, step by step

1. Choroid plexus cells pump Na^+ and other solutes from the plasma into the ventricles; water follows by osmosis.
2. The fluid flows from the two lateral ventricles (the first and second) into the third ventricle.
3. It flows through the cerebral aqueduct, a narrow channel, into the fourth ventricle.
4. It leaves the fourth ventricle and enters the subarachnoid space, between the arachnoid membrane and the pia mater.
5. It circulates around the brain and down around the spinal cord.
6. It is reabsorbed through the arachnoid villi into a venous sinus and returns to the blood. The whole volume turns over about three times a day.

How glucose gets from the blood into a neuron, step by step

1. Glucose arrives in the blood of a brain capillary.
2. Tight junctions close the gaps between the endothelial cells, so it cannot slip between them.
3. A glucose transporter (GLUT1) carries it into the endothelial cell, and another carries it out on the brain side, by facilitated diffusion.
4. Neurons take it up and use it for aerobic metabolism. Astrocytes also take it up and pass lactate to neurons.

Protecting the brain

- Brain and spinal cord tissue is soft, with little extracellular matrix, so it relies on outside support: bone, three membranes, and fluid.
- Gray matter: cell bodies, dendrites and unmyelinated axons. Clusters of cell bodies in the CNS are nuclei; outside the CNS they are ganglia.
- White matter: mostly myelinated axons. Bundles of axons in the CNS are tracts, the CNS equivalent of nerves.

Membrane, skull inward	What it is like	What goes with it
Dura mater	Outermost and thickest	Venous sinuses that drain the brain
Arachnoid membrane	Middle, loosely attached to the pia	The subarachnoid space beneath it, filled with cerebrospinal fluid
Pia mater	Innermost and thin, clings to the brain and cord	Arteries that supply the brain

Cerebrospinal fluid

Choroid plexus

Capillaries covered by a transporting epithelium derived from the ependyma, on the walls of the ventricles. Secretes cerebrospinal fluid.

Physical protection

Buoyancy makes the brain act about 30 times lighter, reducing pressure on its vessels and nerves; the fluid cushions the brain during a blow because water barely compresses.

Chemical protection

The choroid plexus is selective, so the fluid differs from plasma: less K^+ , more H^+ (lower pH), similar Na^+ , very little protein, no blood cells.

- About 150 mL is present at any time; about 500 mL is made each day.
- Lumbar puncture (spinal tap): a sample taken from the subarachnoid space in the lower back, below the end of the spinal cord. Protein or blood cells in it suggest infection.
- Aqueductal stenosis: a blocked cerebral aqueduct makes the lateral and third ventricles enlarge while the fourth stays normal. If all four enlarge, the block is farther along, in the subarachnoid space. Build-up of fluid is hydrocephalus.

The blood-brain barrier

- Most brain capillaries have tight junctions between their endothelial cells, blocking movement between the cells.
- Tight junctions form in response to paracrine factors from astrocyte foot processes and from pericytes, contractile

cells around the capillary that also regulate its blood flow.

- The barrier is selective transport: carriers move nutrients in, other transporters move wastes out, and any water-soluble molecule without a carrier stays out.

Substance	Crosses?	How
Oxygen	Yes, freely	Small and nonpolar: diffuses through the membranes
Small lipid-soluble drugs (anesthetics, older antihistamines)	Yes	Dissolve through the endothelial membranes
Glucose	Yes	Membrane transporters (GLUT1), facilitated diffusion
L-dopa	Yes	Amino acid transporter
Dopamine, large proteins, other water-soluble molecules with no carrier	No	No gaps and no carrier

- Older antihistamines were lipid soluble, crossed the barrier and caused drowsiness; newer ones are less lipid soluble and do not.
- Parkinson's: dopamine cannot cross, so L-dopa is given; neurons convert it to dopamine inside the brain.
- Areas without the barrier: the hypothalamus, whose neurosecretory hormones must enter the portal capillaries to the anterior pituitary, and the vomiting center in the medulla, which monitors the blood for toxins.

The brain's fuel

- The brain uses about one fifth of the body's oxygen at any moment and, by some estimates, about half of its glucose.
- Neurons normally use only glucose for fuel, which is one reason blood glucose is so closely defended.
- Sustained high blood glucose makes barrier cells down-regulate their glucose transporters. If glucose then falls, the brain cannot get enough fast enough: confusion, irritability, slurred speech, and eventually unconsciousness.